Econ 2
Principles of Macroeconomics

Lecture 15: Monetary Policy, Aggregate Demand/Aggregate Supply

Fractional Reserve Banking

- Bank: a firm that specializes in brokering between savers and borrowers, allowing borrowers to go to one firm instead of 1000s

- Banks are only required to keep a fraction of their deposits on hand as reserves
- Lends out the rest
- Creates new money in the form of a deposit in the borrower's account
- Deposit effectively increases the overall money supply
- Governments regulate banks and provide deposit insurance to protect depositors' funds.

Monetary Policy

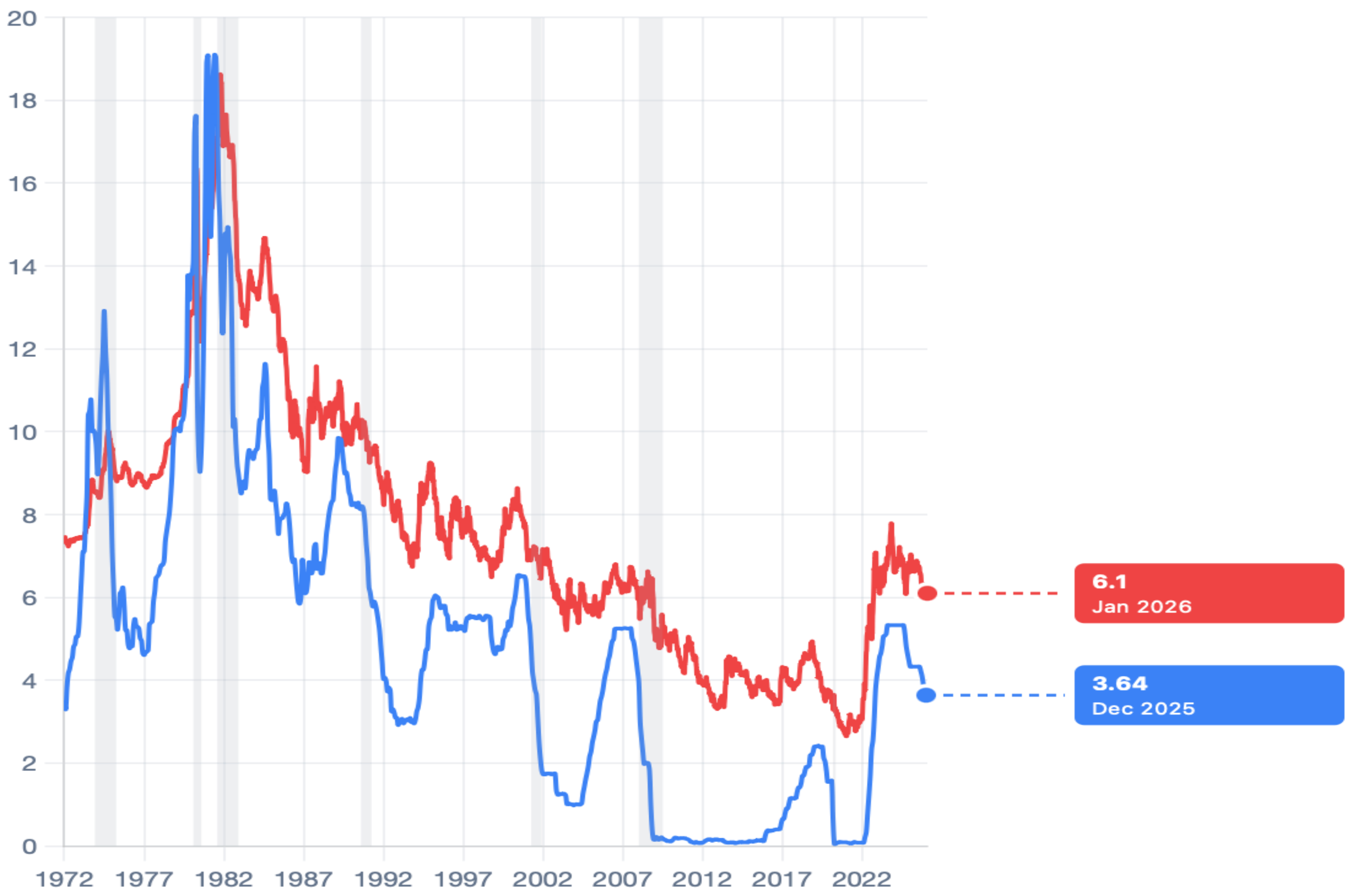
- Buying and selling Treasuries changes the money supply → bond price → interest rate

- Target the Federal Funds Rate: the interest rates banks charge other banks for loans

    - determines flexibility in the banking system since banks can loan out more than they borrow

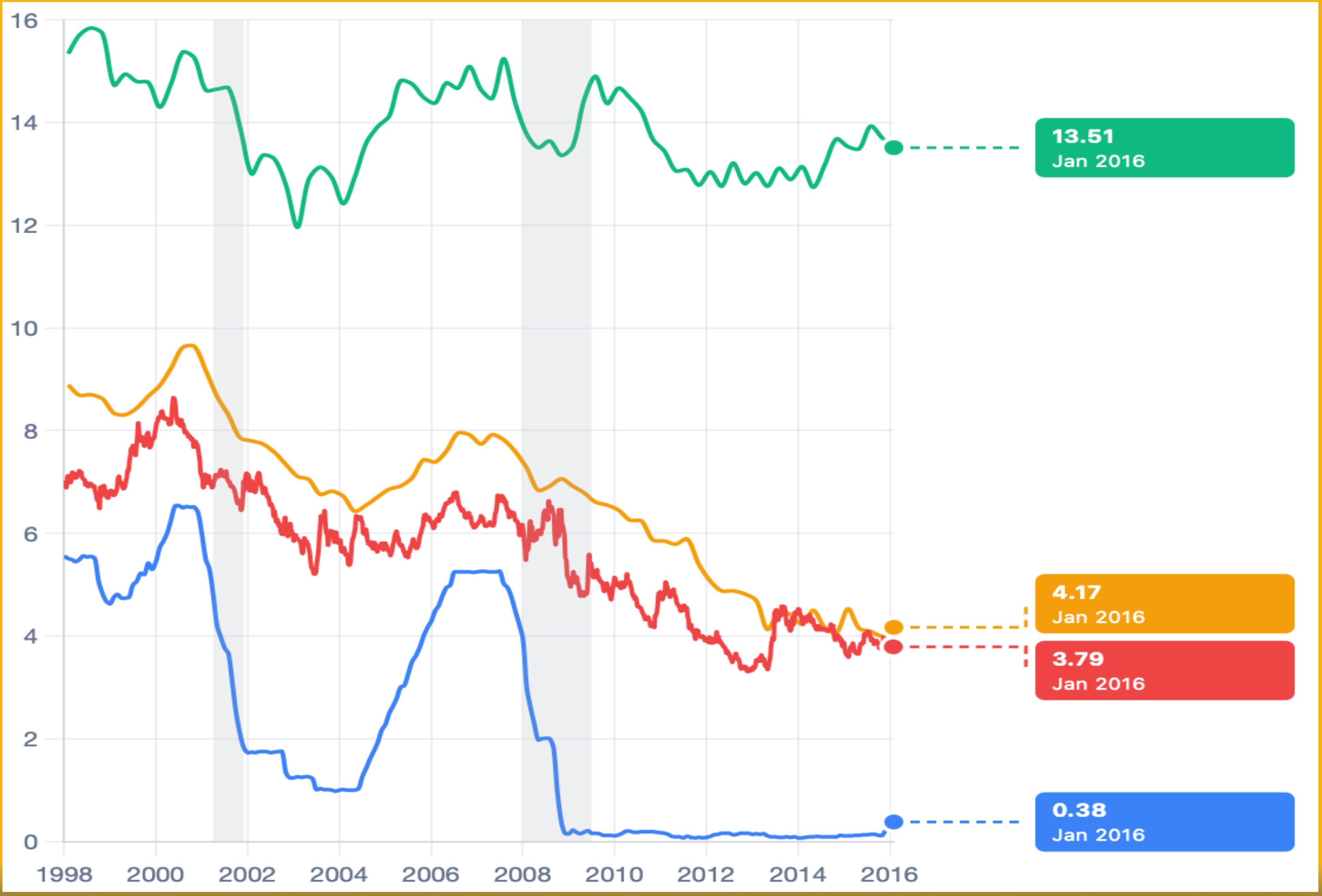
    - other important interest rates tend to follow the Federal Funds rate

Federal Funds Rate vs 30 Year Mort. Rate



Federal Funds Rate vs 30 Year Mortgage Rate, 97-16

Federal Funds Rate vs Interest Rates



Demand for Money

- Connect Monetary Policy to Interest Rates through the market for money
- Demand for Money: Why hold on to money (M1 Money)?

Demand for Money

- Wealth Box:

Money	Other Assets

Demand for Money

- Benefit of holding wealth in the form of money :
- Cost of holding wealth in the form of money:
- Price of money vs. Return on illiquid assets?

Demand for Money

- Interest Rate = r

- Demand for Money:

Demand for Money

Demand for Money

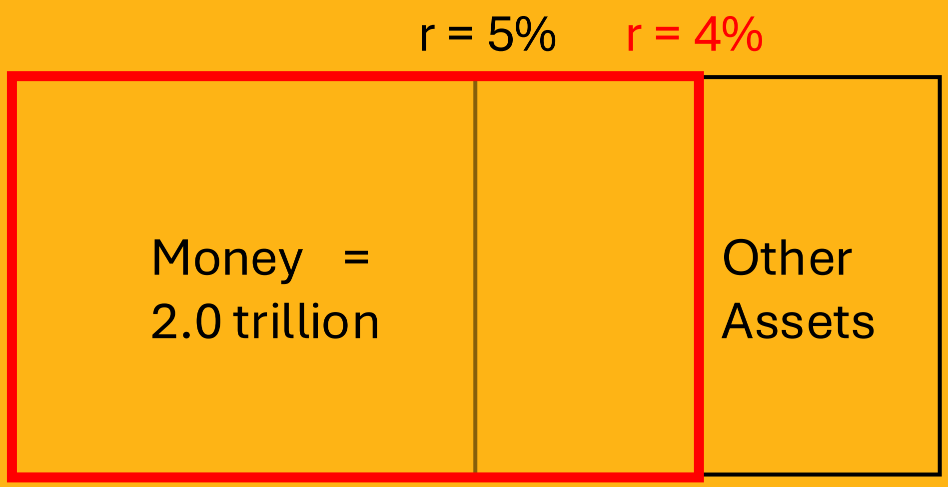
- Point A:

r = 5%

Money 1 trillion	Other Assets

Demand for Money

- Point A to Point B:



Demand for Money

- What shifts the demand for money?

Demand for Money

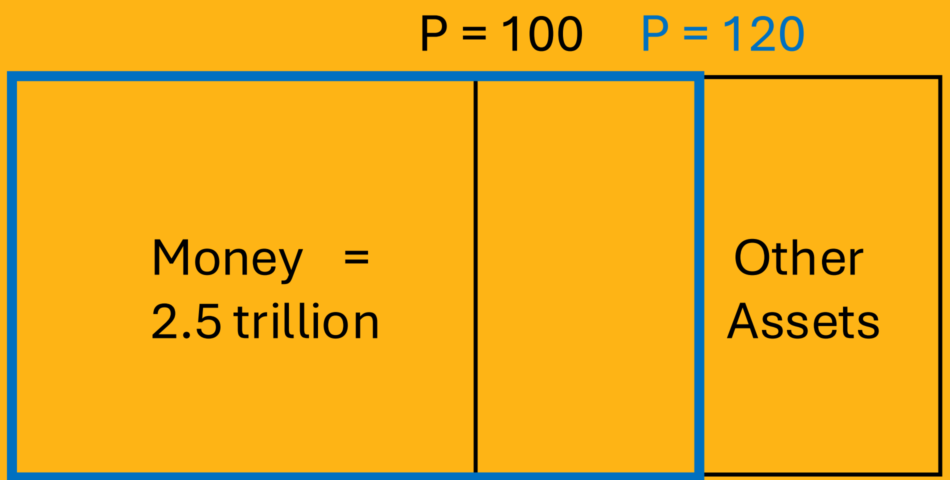
- Point A: r = 5%

P = 100

Money = 1 trillion	Other Assets

Demand for Money

- Point A to Point D:



Demand for Money

- What shifts the demand for money?

Demand for Money

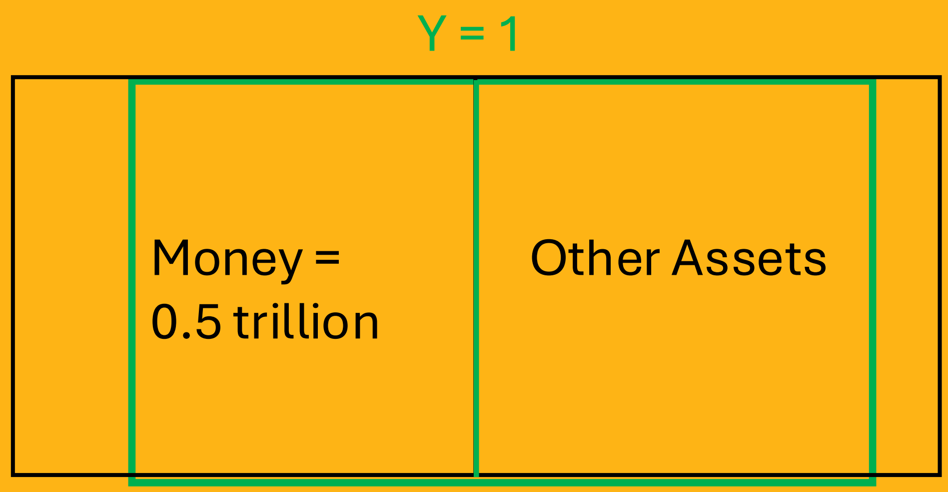
- Point A: r = 5%, P=100

Y = 2

Money = 2 trillion	Other Assets

Demand for Money

- Point A to Point F: r = 5%, P=100



Supply of Money

- Set by the Central Bank/Federal Reserve
- Same supply of money at all interest rates

Money Market Equilibrium

Changing the Interest Rate

- Federal Reserve buys/sells bonds
  - Changes the interest rate

- 1-Year Bond with a value of $100
  - Buyer purchases bond today, gets $100 in 1 year
  - Price of bond today is less than $100

- Interest Rate = (Return on the bond/Price of bond) x 100
- Bond Price Change→ Return on bond Change→ r* Change

Carrying out Monetary Policy

- How does the Fed enact monetary policy?
- Example: r_1 = 5%, Fed wants to decrease r_2 = 4%

Carrying out Monetary Policy

- How does the Fed enact monetary policy?
- Example: r_1 = 5%, Fed wants to increase r_3 = 6%

Why is the interest rate so important?

- What changes when r decreases?
- 1.

- 2.
- 3.

Monetary Policy in Action

- Example 1: Y_1 = 8,000, r_1 = 6% and M_1 = 1 trillion
- $\overline{Y}$ = 10,000 (full employment GDP)

Monetary Policy in Action

- Example 1: Y_3 = 12,000, r_3 = 2% and M_3 = 4 trillion
- $\bar{Y}$ = 10,000 (full employment GDP)

Effects of Inflation

- Example: $\overline{Y}$ = 10,000 (full employment GDP), r_1 = 5%, M_1 = 1 trillion, CPI_1 = 100